

COMPUTER SCIENCE TRIPOS Part IB, Part II 50% – 2022 – Paper 7

6 Further Graphics (aco41)

Implicit functions (a) Which of the following is an implicit function for a closed curve? Briefly explain.

(i) $x^2 + (xy)^2$, $x, y > 0$ [1 mark]

(ii) $e^{f(x,y)} - 1$, where $f(x, y)$ is an implicit function for a closed curve. [1 mark]

(iii) $f(x, y) = 1$ if $g(x, y) > 1$ and $f(x, y) = g(x, y)$ otherwise, where $g(x, y)$ is an implicit function for a closed curve. [1 mark]

(iv) $f(x, y)g(x, y)$, where f and g are implicit functions for circles of radius 2, one centered at $(0, 0)$ and the other at $(1, 0)$. [2 marks]

Answer:

(i) No, this function cannot be 0 anywhere. [1 mark]

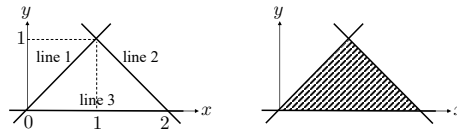
(ii) Yes, it has the same zero set as f . [1 mark]

(iii) Yes, it has the same zero set as g . [1 mark]

(iv) No, this represents the intersection of two circles, which is not a closed curve. [2 marks]

Representing a triangle with implicit functions

(b) In this question, we will derive an implicit representation for a triangle.



(i) Write the implicit function for a line passing through $(0, 0)$ on the xy -plane. [1 mark]

(ii) Derive the implicit functions of the three lines in the figure on the left. [3 marks]

(iii) Derive an implicit function representing the triangle that is formed by the three lines. The function is 0 inside the triangle (shaded in the figure) and non-zero otherwise. [Hint: You may use the function $\max(0, x)$.] [3 marks]

Answer:

(i) $ax + by$. [1 mark]

(ii) Line 1: $f_1(x, y) = -x + y$, line 2: $f_2(x, y) = x + y - 2$, line 3: $f_3(x, y) = -y$, up to sign. [3 marks]

(iii) $t(x, y) = m(f_1(x, y)) + m(f_2(x, y)) + m(f_3(x, y))$, where $m(x) = \max(0, x)$ [3 marks]

Quaternions

(c) In this question, we represent rotations in the xy -plane with quaternions.

(i) Write the quaternion representing a rotation of angle θ around the z -axis. [1 mark]

- (ii) Derive the quaternion for rotation by θ_1 and then by θ_2 around the z -axis.
 [Hint: $\cos(a)\cos(b) - \sin(a)\sin(b) = \cos(a+b)$, and $\sin(a)\cos(b) + \cos(a)\sin(b) = \sin(a+b)$.] [3 marks]
- (iii) Starting from spherical blending of quaternions, prove that shortest path interpolation from the first to the second quaternion above (with $\theta_2 > \theta_1$) is given by: $\mathbf{q}(t) = \cos([(1-t)\theta_1 + t\theta_2]/2) + \hat{z} \sin([(1-t)\theta_1 + t\theta_2]/2)$, where $t \in [0, 1]$. [Hint: Recall that $\mathbf{q}^t = e^{t \log \mathbf{q}}$, $\log \mathbf{q} = \frac{\theta}{2} \mathbf{s}$, and $e^{\mathbf{q}} = \cos \|\mathbf{q}\| + \frac{\mathbf{q}}{\|\mathbf{q}\|} \sin \|\mathbf{q}\|$ for a quaternion $\mathbf{q} = \cos(\theta/2) + \mathbf{s} \sin(\theta/2)$.] [4 marks]

Answer:

- (i) $\mathbf{q} = \cos(\theta/2) + \hat{z} \sin(\theta/2)$. [1 mark]
- (ii) Denote $\mathbf{q}_i = c_i + \hat{z} s_i$, where $i = 1, 2$ and $c_i = \cos(\theta_i/2)$, $s_i = \sin(\theta_i/2)$. If we multiply the two quaternions:

$$\mathbf{q}_1 \mathbf{q}_2 = (c_1 + \hat{z} s_1)(c_2 + \hat{z} s_2) = c_1 c_2 - s_1 s_2 + \hat{z}(s_1 c_2 + c_1 s_2) = \cos((\theta_1 + \theta_2)/2) + \hat{z} \sin((\theta_1 + \theta_2)/2)$$

[3 marks]

- (iii) Spherical blending is given by $(\mathbf{q}_2 \mathbf{q}_1^*)^t \mathbf{q}_1$.

$$\mathbf{q}^t = e^{t \log \mathbf{q}} = e^{t \frac{\theta}{2} \mathbf{s}} = \cos \left\| t \frac{\theta}{2} \mathbf{s} \right\| + \mathbf{s} \sin \left\| t \frac{\theta}{2} \mathbf{s} \right\| = \cos |t\theta/2| + \mathbf{s} \sin |t\theta/2|$$

$$\mathbf{q}_2 \mathbf{q}_1^* = (c_2 + \hat{z} s_2)(c_1 - \hat{z} s_1) = (c_1 c_2 + s_1 s_2) + \hat{z}(s_2 c_1 - c_2 s_1) = \cos((\theta_2 - \theta_1)/2) + \hat{z} \sin((\theta_2 - \theta_1)/2)$$

$$(\mathbf{q}_2 \mathbf{q}_1^*)^t = \cos(t(\theta_2 - \theta_1)/2) + \hat{z} \sin(t(\theta_2 - \theta_1)/2)$$

$$(\mathbf{q}_2 \mathbf{q}_1^*)^t \mathbf{q}_1 = \cos([t(\theta_2 - \theta_1) + \theta_1]/2) + \hat{z} \sin([t(\theta_2 - \theta_1) + \theta_1]/2).$$

[4 marks]